

AI RESUME ANALYZER AND JOB RECOMMENDATION SYSTEM

GitHub Repository: https://github.com/ankithakr-2005/ai_resume_analyzer

ABSTRACT & OBJECTIVE

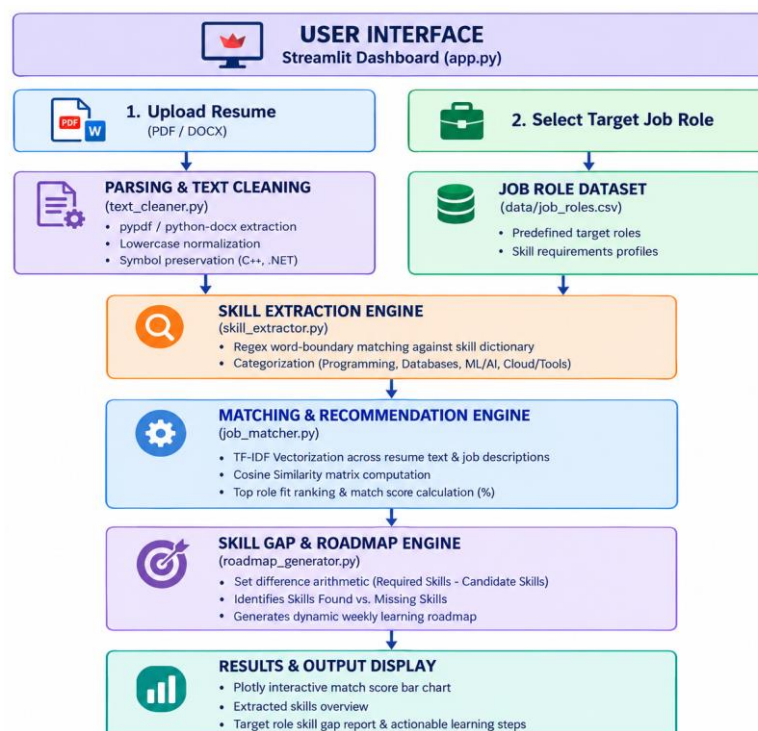
The **AI Resume Analyzer & Job Recommendation System** is an end-to-end Natural Language Processing (NLP) application designed to assist job seekers and recruiters in evaluating resume alignment against targeted industry job roles. The system parses unstructured resume documents in PDF and DOCX formats, cleans and normalizes text, extracts technical skills, and evaluates similarity against defined job profiles. Using **TF-IDF (Term Frequency-Inverse Document Frequency)** vectorization and **Cosine Similarity**, the application calculates percentage match scores, performs a skill gap analysis, and generates a structured, actionable learning roadmap.

PROBLEM STATEMENT & NEED

Job candidates often struggle to determine whether their resumes effectively capture the technical skills required for target job roles. Simultaneously, recruiters require automated tools to parse and rank candidates fairly based on objective qualifications. This project addresses these challenges by offering an educational matching engine that focuses purely on skills, experience, and educational alignment while eliminating bias related to personal attributes.

3. SYSTEM ARCHITECTURE & WORKFLOW

The application follows a modular architecture consisting of five core processing pipeline stages:



Module Descriptions:

1. **Module 1 - Resume Parsing** (text_cleaner.py): Extracts raw text from PDF files using pypdf and DOCX files using python-docx. It cleans whitespace and normalizes text while preserving critical technical terms containing special characters (e.g., C++, C#, .NET).
2. **Module 2 - Skill Extraction Engine** (skill_extractor.py): Utilizes regex boundary matching against a predefined skill taxonomy to accurately identify candidate technical skills across categories (Programming, Databases, ML/AI, Cloud/Tools).
3. **Module 3 - Dataset Management** (data/job_roles.csv): Houses structured skill profiles for key industry roles, including *Data Analyst*, *Machine Learning Engineer*, *AI Engineer*, *NLP Engineer*, and *Computer Vision Engineer*.
4. **Module 4 - Matching & Recommendation** (job_matcher.py): Transforms clean text into vector representations using scikit-learn's TF-IDF Vectorizer and computes Cosine Similarity match percentages across all roles.
5. **Module 5 - Skill Gap & Roadmap Generator** (roadmap_generator.py): Executes set-difference operations between target role requirements and candidate skills to identify missing skills and construct weekly learning recommendations.
6. **Module 6 - Streamlit Dashboard** (app.py): Provides an intuitive user interface with file upload capabilities, interactive Plotly match bar charts, and downloadable analysis summaries.

KEY FEATURES & CAPABILITIES

- **Multi-Format Support:** Seamlessly parses PDF and DOCX resume formats.
- **Symbol-Preserving Preprocessing:** Retains technical syntax (e.g., C++, C#, .NET) to prevent extraction loss.
- **Vector-Based Match Scoring:** Uses TF-IDF and Cosine Similarity to score resume relevance mathematically.
- **Target Role Skill Gap Analysis:** Highlights both matching skills and missing skill requirements.
- **Actionable Roadmaps:** Provides weekly step-by-step study plans based on missing competencies.

5. TESTING AND EVALUATION RESULTS

The pipeline was evaluated across anonymized test resumes representing various candidate technical domains:

Test Case	Uploaded Format	Target Role	Top Recommendation	Match Score (%)	Status
Resume	PDF	Data	Data Analyst	86.0%	PASS

Test Case	Uploaded Format	Target Role	Top Recommendation	Match Score (%)	Status
A		Analyst			
Resume B	PDF	ML Engineer	Machine Learning Engineer	74.0%	PASS
Resume C	DOCX	NLP Engineer	NLP Engineer	81.5%	PASS

RESPONSIBLE AI & ETHICAL COMPLIANCE

- **Bias Reduction:** The model scores resumes purely on technical skills, education, and relevant experience. Personal attributes such as age, gender, ethnicity, address, or photos are strictly excluded.
- **Data Privacy:** Files are parsed dynamically in temporary memory and are deleted post-analysis.

CONCLUSION & FUTURE SCOPE

The **AI Resume Analyzer** provides an effective, objective tool for resume evaluation and career progression. Future improvements include implementing semantic embeddings via **Sentence Transformers**, integrating LLM-driven feedback for resume formatting, and adding direct job posting scraper integrations.